



## **SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY**

An Autonomous Institution | Approved by AICTE | Affiliated to Anna University | Accredited by NAAC with A++ Grade  
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### **INTERNSHIP REPORT**

**Submitted by**

**MOHANA PRASANTH S (727724EUCB501)**

**in partial fulfilment of the requirements for the award of the degree of**

**BACHELOR OF TECHNOLOGY**

***in***

**COMPUTER SCIENCE AND BUSINESS SYSTEMS**

**August 2026**



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### **DEPARTMENT OF COMPUTER SCIENCE AND BUSINESS SYSTEMS**

### **BONAFIDE CERTIFICATE**

Certified that this **INTERNSHIP** report is the Bonafide work of  
**MOHANA PRASANTH S (727724EUCB501)** who carried out **INTERNSHIP** at  
**BI3 TECHNOLOGIES** from 01.07.2026 to 01.09.2026.

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### SUSTAINABLE DEVELOPMENT GOALS (SDGs)

The Sustainable Development Goals are a collection of 17 global goals designed as a blueprint to achieve a better and more sustainable future for all. The SDGs, set in 2015 by the United Nations General Assembly and intended to be achieved by the year 2030, reflect an agreement among 195 nations to work toward a better world. The project is based on one of the 17 goals.

| Questions                                                                     | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Which SDGs does the project directly address?                                 | SDG 9 - Industry, Innovation and Infrastructure, SDG 8 - Decent Work and Economic Growth, and SDG 4 - Quality Education.                                                                                                                                                                                                                                                                                                                                  |
| What strategies or actions are being implemented to achieve these goals?      | The Saluda project builds a layered enterprise data platform that consolidates information from multiple business source systems into a single, governed environment. By organising data through Bronze, Silver, Silver Conformed and Gold layers, the platform creates reliable digital infrastructure that supports informed decision-making, reduces manual and duplicated data handling, and makes trustworthy information available to the business. |
| How is progress measured and reported in relation to the SDGs?                | Progress is assessed through the completeness and correctness of the data loaded into each layer, the outcome of unit testing performed on Bronze and Silver layer data, the successful integration of the three source systems, the standardisation achieved in the Silver Conformed layer, and the availability of governed, business-ready data in the Gold layer.                                                                                     |
| How were these goals identified as relevant to the project's objectives?      | These goals were identified because the project develops modern digital infrastructure (SDG 9), strengthens the data foundation that organisations depend on for productivity and sustained economic activity (SDG 8), and provides structured, practical learning in data engineering to an intern through industry-based training (SDG 4).                                                                                                              |
| Are there any partnerships or collaborations in place to enhance this impact? | The internship itself represents a collaboration between Sri Krishna College of Engineering and Technology and BI3 Technologies. Within the project, work is carried out in collaboration with the project team, source system owners,                                                                                                                                                                                                                    |

## CERTIFICATE FROM THE INDUSTRY

The internship offer issued by BI3 Technologies is placed below.



22-June-2026

Dear Mohana Prasanth,

Congratulations!

We are pleased to offer you an opportunity to join Bi3 Technologies as a "**Data Engineer - Trainee**", based on the successful outcome of your interviews with us. Your annual compensation package will be **INR,4,50,636**, inclusive of a variable pay component.

**\*\*TDS** will be applicable as per Government norms.

We are confident that your knowledge, skills, and potential will be a valuable addition to our organization, and we look forward to your contribution to the team.

During the internship period, you will receive a monthly stipend of **INR 15,000**. Upon successful completion of the internship and based on satisfactory performance, you will be considered for confirmation as a full-time employee following a performance evaluation.

Your formal appointment letter will be issued on your Date of Joining to start your **internship**. We request you to join us on **1 July 2026**.

If you decide to accept this offer, kindly send an acknowledgement email to [pradeepa.sivakumar@bi3technologies.com/hr@bi3technologies.com](mailto:pradeepa.sivakumar@bi3technologies.com/hr@bi3technologies.com) no later than the **24-June-2026**. Please do not hesitate to reach out to us if you have any queries or require further assistance.

**Please report to Bi3 Technologies India Pvt Ltd (Coimbatore/Chennai) at 9:30 am on your joining date.**

**Note:** The reporting location will be communicated to you in advance of your joining date.

| Certification Field    | Official Record Detail                       |
|------------------------|----------------------------------------------|
| Student Candidate Name | Mohana Prasanth S                            |
| Register Number        | 727724EUCB501                                |
| Issuing Entity         | BI3 Technologies                             |
| Role                   | Data Engineer Intern                         |
| Project                | Saluda – Data Platform Building              |
| Internship Duration    | 01 July 2026 to 01 September 2026 (2 months) |

## ABSTRACT

Modern organisations generate and store business information across several independent software systems, each designed for a specific operational purpose. When this information remains isolated within those individual systems, it becomes difficult to obtain a single, consistent and trustworthy view of the business. Reports produced from different systems may disagree with one another, the same business entity may be represented differently in each system, and analytical teams are forced to spend significant effort reconciling data before it can be used. The Saluda project addresses this problem by building an enterprise data platform that brings information from multiple source systems into a common, governed environment where it can be progressively cleaned, standardised and prepared for business use.

The Saluda platform draws data from three source systems – QAD, SharePoint and Salesforce – and passes it through a layered architecture consisting of a Bronze layer, a Silver layer, a Silver Conformed layer and a Gold layer. Each layer performs a defined role: the Bronze layer preserves the ingested source data, the Silver layer applies cleaning and transformation, the Silver Conformed layer standardises and integrates data originating from different systems into consistent business entities, and the Gold layer presents refined, business-oriented data for downstream analytical consumption. This layered approach allows data quality to be improved in controlled stages and makes it possible to trace any value in the final output back to its origin in a source system.

My internship at BI3 Technologies was carried out over a period of two months, from 01 July 2026 to 01 September 2026, in the role of Data Engineer Intern. The first three weeks consisted of structured technical training in SQL, Snowflake and Databricks, which established the foundation required to work with structured data, cloud data warehousing and modern data engineering platforms. From the fourth week onwards I was assigned to the Saluda project, where I began by studying the problem being addressed and analysing the three source systems and the data they contribute to the platform. In the following weeks I performed unit testing of Bronze and Silver layer data for all three source systems, contributed to data modelling work for the Silver Conformed layer, worked on security governance for the Bronze and Silver layers, and supported the movement of data from the Silver layer towards the Silver Conformed and Gold layers.

Beyond the technical exposure, the internship provided practical understanding of how enterprise data platforms are designed, validated, governed and delivered in a professional environment. It reinforced the importance of verifying data before it is trusted, of standardising information originating from heterogeneous systems, and of controlling access to data at every layer. The work also aligns with SDG 9 - Industry, Innovation and Infrastructure, SDG 8 - Decent Work and Economic Growth, and SDG 4 - Quality Education, through its contribution to reliable digital infrastructure, organisational productivity, and industry-based practical learning.



## PICTURES CAPTURED DURING INTERNSHIP

Photographs and screenshots captured during the internship period at BI3 Technologies are placed below.



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# CHAPTER 1

## INTRODUCTION

Organisations today operate using several separate software systems, each built to support a particular business function and each storing data in its own structure, with its own naming conventions and definitions of business terms. Individually each system performs its intended function well; collectively they produce a fragmented picture of the organisation. A data platform is a structured environment in which data from such systems is collected, processed, standardised and made available for analysis and reporting. Building one is the responsibility of data engineering, the discipline concerned with the design, construction, validation and governance of systems that move and prepare data.

The Saluda project is a data platform building project undertaken at BI3 Technologies. The platform integrates data from three source systems – QAD, SharePoint and Salesforce – and processes it through a layered architecture: data is landed in a Bronze layer, cleaned in a Silver layer, standardised and integrated in a Silver Conformed layer, and refined into business-oriented datasets in a Gold layer. Each layer increases the reliability of the data while preserving the ability to trace it back to its origin.

This report describes the work I carried out as a Data Engineer Intern at BI3 Technologies between 01 July 2026 and 01 September 2026: a training phase covering SQL, Snowflake and Databricks, followed by a project phase contributing to the Saluda platform.

### 1.1 Problem Statement

When business data remains confined to the individual systems that produce it, the organisation loses the ability to view its operations as a whole. The same customer, product or transaction may be recorded differently in each system, and combining such data manually is slow and prone to error. The major problems addressed by the project include:

- Business data is distributed across multiple independent source systems, each using its own data structures, field names and business definitions.
- There is no single trusted location where integrated organisational data is available.
- Uncontrolled access to raw and intermediate data creates security and governance risks.

There is therefore a need for a structured, layered and governed data platform that consolidates data from multiple source systems, validates it, standardises it and delivers it in a form suitable for business consumption.

### 1.2 Purpose of the Project

The purpose of the Saluda project is to build a data platform that brings together data from the QAD, SharePoint and Salesforce source systems and processes it through defined layers, so that the organisation has access to consistent and reliable information. Separating the concerns of ingestion, cleaning, standardisation and business presentation across layers makes the platform



easier to test, govern and extend. A further purpose is to ensure that data is not simply moved but verified: unit testing at the Bronze and Silver layers confirms that data has been transferred and transformed correctly, while security governance ensures that it is accessible only to those authorised to view it.

### **1.3 Need for the Project**

As organisations increase their reliance on data for planning and decision-making, a reliable and well-governed data foundation becomes essential. A data platform is needed because data from multiple systems must be integrated to obtain a complete organisational view; because decisions require data that has been validated rather than assumed correct; because consistent definitions of business entities are needed across departments; and because data must be traceable, and access to it controlled, at every stage. The Saluda project therefore acts as the intermediary between the source systems that generate data and the business functions that consume it.

## **CHAPTER 2**

### **INDUSTRY PROFILE**

#### **2.1 Data Engineering Industry Overview**

The data engineering industry has expanded rapidly as organisations have increased their dependence on data-driven decision-making. Data engineering is the discipline concerned with designing and building the systems through which data moves: extraction from source systems, storage, transformation into usable forms, validation of quality and delivery to analytical layers. Where data analysis focuses on interpreting data, data engineering ensures that the data being interpreted is complete, correct and available.

The industry has moved steadily from on-premises databases towards cloud-based platforms, which allow organisations to store and process large volumes of data without maintaining physical infrastructure. Alongside this shift, increasing emphasis has been placed on data quality, governance and security, which are now established parts of data engineering practice rather than optional additions.

#### **2.2 Modern Data Platforms**

A modern data platform is an organised environment in which data from multiple sources is ingested, processed and delivered in a controlled sequence. Rather than transforming raw data into final reports in a single step, it processes data through successive layers, each producing data of a higher standard than the one before it. This approach is commonly described as a layered or medallion architecture, typically consisting of:

- A Bronze layer, which holds data as received from the source systems, in its original form.
- A Silver layer, in which data is cleaned, validated and transformed into a usable state.
- A conformed layer, in which data from different sources is standardised into consistent business entities.
- A Gold layer, which contains refined, business-oriented data prepared for reporting and analytics.

Because each layer has a clearly defined responsibility, problems can be identified and corrected at the stage where they occur; because the raw data is preserved, processing can be repeated without returning to the source systems; and because the layers are separated, testing, governance and access control can be applied appropriately to each stage.

#### **2.3 BI3 Technologies Profile**

BI3 Technologies is a technology services organisation working in the area of data and analytics, delivering data platform and data engineering solutions for client organisations. The Saluda project, on which I worked during my internship, is one such engagement. The organisation provides a professional working environment in which interns receive structured technical training before being assigned to project work.

| Particulars                  | Details                                                                   |
|------------------------------|---------------------------------------------------------------------------|
| Organisation Name            | BI3 Technologies                                                          |
| Nature of Business           | Data and analytics services; data platform and data engineering solutions |
| Internship Role              | Data Engineer Intern                                                      |
| Internship Domain            | Data Engineering / Data Platform Building                                 |
| Project Assigned             | Saluda – Data Platform Building                                           |
| Internship Duration          | 01 July 2026 to 01 September 2026 (2 months)                              |
| Registered Office / Location | Coimbatore , Indicube                                                     |
| Year of Establishment        | 2021                                                                      |
| Website                      | <a href="#">BI3 Technologies — Enterprise Data and AI Consulting</a>      |
| Industry Mentor / Supervisor | Pradeepa -HR                                                              |

## 2.4 Importance of Data Engineering

Data engineering is important because every analytical output an organisation produces depends on the quality of the data beneath it. A dashboard, a forecast or a management report is only as reliable as the pipeline that supplied its data: if data is incomplete, duplicated, inconsistently defined or incorrectly transformed, the conclusions drawn from it will be wrong regardless of the sophistication of the analysis applied. Its importance can be summarised as follows:

- It creates a single consistent source of organisational data from otherwise isolated systems.
- It ensures that data is validated before it is used for business purposes.
- It establishes standard definitions of business entities across departments.
- It provides traceability, linking any reported value back to its source, and applies security and governance controls at every stage of processing.

## **CHAPTER 3**

### **OBJECTIVES OF THE PROJECT**

#### **3.1 Primary Objective**

The primary objective of the Saluda project is to build a layered enterprise data platform that consolidates data from the QAD, SharePoint and Salesforce source systems and processes it through Bronze, Silver, Silver Conformed and Gold layers, so that validated, standardised and business-ready data is made available for downstream consumption.

The primary objective of my internship was to develop practical data engineering capability and apply it to the Saluda project by analysing source systems, validating processed data through unit testing, contributing to data modelling for the Silver Conformed layer, and supporting security governance and the movement of data towards the higher layers of the platform.

#### **3.2 Secondary Objectives**

1. To acquire a working knowledge of SQL for querying, examining and validating structured data.
2. To understand cloud data warehousing concepts through training in Snowflake.
3. To understand the role of Databricks in modern data engineering and data processing.
4. To study the Saluda project and the business problem the data platform addresses.
5. To analyse the QAD, SharePoint and Salesforce source systems and the data each contributes.
6. To perform unit testing of Bronze and Silver layer data for all three source systems.
7. To contribute to data modelling work for the Silver Conformed layer.
8. To work on security governance for the Bronze and Silver layers.
9. To support the movement of data from the Silver layer towards the Silver Conformed and Gold layers.
10. To develop professional working skills through participation in a live project team.

### **3.3 Measurable Outcomes**

The outcomes of the internship can be assessed against the following measures:

- Completion of the structured training programme in SQL, Snowflake and Databricks.
- Completed analysis of the three source systems and the data they contribute.
- Unit testing completed for Bronze and Silver layer data across QAD, SharePoint and Salesforce.
- Contribution delivered towards Silver Conformed layer data modelling.
- Security governance work completed for the Bronze and Silver layers.
- Support provided for the movement of data towards the Silver Conformed and Gold layers.
- Consistent participation, task completion and professional conduct across the eight-week period.

Quantitative measures such as record counts, defect counts and processing times were maintained within the project environment and are not reproduced in this report, as project data is confidential to the organisation and its client.

## **CHAPTER 4**

### **SCOPE OF THE PROJECT**

#### **4.1 Source System Analysis**

Source system analysis is the activity of studying the systems that supply data to a platform before any transformation or modelling work is designed. It involves understanding what each system is used for, what data it holds, how that data is structured, how frequently it changes, and how it relates to data held in other systems.

This analysis is a prerequisite for every later stage: transformations cannot be written correctly without knowing the structure and meaning of the incoming data, data models cannot be designed without knowing which business entities exist in which systems, and validation rules cannot be defined without knowing what a correct value looks like in the source. For the Saluda platform, three source systems were analysed: QAD, SharePoint and Salesforce.

#### **4.2 QAD**

QAD is an enterprise resource planning (ERP) system. ERP systems are used to manage core operational functions and typically hold structured transactional and master data relating to those functions. Within the Saluda platform, QAD is one of the three systems supplying data to the Bronze layer. As an ERP source its data is generally well structured and relational, and can therefore be examined and validated using SQL-based querying. I studied the QAD data brought into the platform during source system analysis and validated it at the Bronze and Silver layers during testing. The specific modules, tables and fields extracted are internal to the project.

#### **4.3 SharePoint**

SharePoint is a collaboration and document management platform used to store documents, maintain lists and share information across teams. As a source system it differs in character from an ERP system: data held in a collaboration platform is often maintained by business users rather than generated by automated transactional processes, and may therefore vary more in structure and completeness. Recognising this is important when designing validation rules, because expectations reasonable for ERP data may not hold for manually maintained data. SharePoint is the second source system contributing to the Bronze layer, and I analysed and validated its data during the internship.

#### **4.4 Salesforce**

Salesforce is a customer relationship management (CRM) platform used to manage information relating to customers, accounts, contacts and commercial activity. Because entities such as customers and accounts may also be represented in other systems, Salesforce data is particularly relevant to the conformed layer, where representations of the same business entity originating from different systems must be reconciled into a single standardised definition. It is the third source



system contributing to the Bronze layer and was included in my source system analysis and in the Bronze and Silver layer unit testing.

#### **4.5 Data Quality and Testing**

Data quality and testing form a defined part of the platform's scope rather than an activity performed only at the end. Data moved from a source system into a platform layer cannot be assumed correct simply because the process completed without error; a process can run successfully while still producing incomplete or incorrect results. The testing scope includes source-to-target validation, confirming that data present in the source is correctly represented in the target layer; completeness checks, confirming that no expected records or values are missing; consistency and correctness checks; and transformation validation, confirming that logic applied between layers produced the intended result. Within this scope, I performed unit testing of Bronze and Silver layer data for all three source systems.

#### **4.6 Data Modelling**

Data modelling is the activity of defining how data is organised, how business entities are represented and how those entities relate to one another. Within the Saluda platform it is particularly significant at the Silver Conformed layer, where data originating from three different source systems must be brought into a common structure. Each source system represents business information according to its own internal design; the conformed model defines a single agreed representation that all three can be mapped into. My contribution within this scope was to work on data modelling for the Silver Conformed layer during Week 6. The specific entities, structures and relationships defined within the project model are confidential and are not reproduced here.

#### **4.7 Security Governance**

Security governance defines who may access data, at which layer, and under what conditions. It falls within the platform's scope because a data platform concentrates organisational data into a single environment, which increases both its value and the consequences of uncontrolled access. The scope includes controlling access to data held in each layer, recognising that different layers carry different sensitivity and require different restrictions, and protecting raw and intermediate data that has not been curated for general consumption. Within this scope, I worked on security governance for the Bronze and Silver layers during Week 7.

#### **4.8 Future Scope**

The platform and the work carried out may be extended in the following directions:

- Extension of unit testing coverage to the Silver Conformed and Gold layers.
- Automation of data quality checks so that validation is executed routinely rather than manually.
- Integration of additional source systems into the existing layered architecture.
- Extension of security governance controls to the Silver Conformed and Gold layers.

## **CHAPTER 5**

### **ORGANIZATION / PROJECT STRUCTURE**

#### **5.1 Saluda Project Structure**

The Saluda project is structured as a layered data platform. Data enters from three source systems and progresses through a defined sequence of layers, each performing a specific function and producing data of a higher standard than the layer preceding it. The structure divides into the following functional areas: the source systems (QAD, SharePoint and Salesforce) which supply data; the Bronze layer, in which source data is held in its ingested form; the Silver layer, in which data is cleaned, transformed and structured; the Silver Conformed layer, in which data from all sources is standardised and integrated; and the Gold layer, which contains refined, business-oriented data for downstream consumption. Running across these layers are two further concerns: data quality and testing, which validates the processed data, and security and governance, which controls access to it.

This separation of responsibilities is the defining characteristic of the structure. Because each layer has a single defined purpose, work can be developed, tested and governed independently at each stage, and issues can be traced to the layer in which they originate.

#### **5.2 Source Systems**

Three source systems supply data to the Saluda platform: QAD, an enterprise resource planning system contributing structured operational data; SharePoint, a collaboration and document management platform contributing information maintained by business users; and Salesforce, a customer relationship management platform contributing customer and commercial data.

These systems are independent of one another. Each was designed for its own purpose, and none was designed with the others in mind. Consequently the same real-world business entity may appear in more than one system, described in different ways and identified by different values. This is precisely the condition that the platform, and in particular the conformed layer, exists to resolve, and it is the reason that analysis of the source systems precedes the design of transformations and models.

#### **5.3 Bronze Layer**

The Bronze layer is the landing area for data arriving from the source systems. Its purpose is to receive and retain source data in its ingested form, without applying business transformation. Retaining data in this state provides a record of what was actually received, allows processing to be re-run without extracting from the source systems again, provides a reference point for validating transformed data, and establishes the starting point for tracing lineage. Because the data is raw and uncurated it is not intended for general business consumption, which is one reason access to this layer requires particular governance attention. During Week 5 I performed unit testing of Bronze layer data for all three source systems, and during Week 7 I worked on security governance for this layer.

## 5.4 Silver Layer

The Silver layer receives data from the Bronze layer and applies cleaning and transformation, converting raw ingested data into a structured, usable and reliable form. At this stage data from each source system has improved in quality but is still organised largely according to the characteristics of the system it came from; it exists in cleaned form but has not yet been reconciled into shared business definitions, which is the function of the next layer. During Week 5 I performed unit testing of Silver layer data for all three source systems, and during Week 7 I worked on security governance for this layer.

## 5.5 Silver Conformed Layer

The Silver Conformed layer sits between the Silver and Gold layers and standardises and integrates data originating from different source systems, so that a single consistent representation of each business entity exists within the platform. It is required because cleaning data does not by itself make data from different systems comparable: three systems may each hold valid data about the same entity while describing it in three different ways, and combining such data without standardising it first would produce results that appear integrated but are not genuinely consistent.

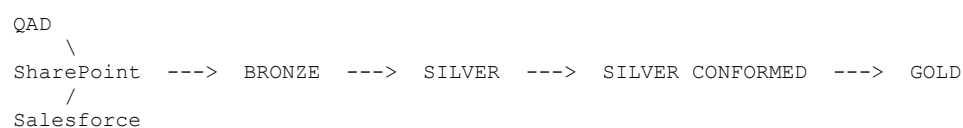
Data modelling is central to this layer, since the standardised representation must be designed before data can be mapped into it. During Week 6 I was assigned to work on data modelling for this layer, and during Week 8 I worked on the movement of data from the Silver layer towards it.

## 5.6 Gold Layer

The Gold layer is the final layer and contains refined, business-oriented data prepared for downstream consumption. Data reaching it has been ingested, cleaned, validated, standardised and integrated, and is organised according to business meaning rather than the structure of the systems it originated from. Its importance lies in what it removes from the consumer: users need not know which source system supplied a value, how the underlying systems differ, or what transformation was applied. During Week 8 I worked on the movement of data from the Silver layer towards the Silver Conformed and Gold layers.

## 5.7 Overall Data Flow

The overall structure of the Saluda platform can be represented conceptually as follows. Each layer depends on the one before it, so the quality of the final output is determined by every preceding stage – which is why validation is applied early, at the Bronze and Silver layers, rather than only at the point of consumption.



*Figure 1 – Overall Saluda Data Platform Architecture*

## **CHAPTER 6**

### **TOOLS AND TECHNOLOGIES USED**

#### **6.1 SQL**

SQL (Structured Query Language) is the standard language used to query and manipulate data held in relational and structured data stores. It was the first subject of my training during Week 1, covering fundamentals and practical querying together with working with structured data and performing data querying and validation. Within data engineering, SQL serves two related purposes: expressing the logic by which data is transformed as it moves between layers, and examining and verifying data once processed. The second was directly relevant to my project work, since the unit testing I performed during Week 5 depended on the ability to query data and compare a target layer against what was expected from the source.

#### **6.2 Snowflake**

Snowflake is a cloud data warehousing platform. A data warehouse is a system designed to store large volumes of structured data and support analytical querying over it; a cloud data warehouse provides this capability as a managed service, without the organisation maintaining the underlying physical infrastructure. Training in Snowflake began in Week 1 alongside SQL and continued through Week 2, covering Snowflake concepts, cloud data warehousing, data storage and processing, SQL-based data operations, and how Snowflake can be used within modern data platforms.

#### **6.3 Databricks**

Databricks is a platform used for data engineering and data processing, providing an environment in which data can be processed and transformed at scale as part of a data pipeline. I received Databricks training during Week 3, covering an introduction to the platform, modern data engineering, data processing, data transformation, and the role of Databricks within data platforms. Where SQL and Snowflake addressed structured querying and cloud data warehousing, Databricks addressed the processing and transformation of data as it moves between layers.

#### **6.4 Data Modelling**

Data modelling is the practice of defining how data is organised within a system: which business entities are represented, what attributes describe them, and how they relate to one another. Within the Saluda platform it is applied most significantly at the Silver Conformed layer, where a common structure must be defined so that data from three different source systems can be represented consistently. I contributed to Silver Conformed layer data modelling during Week 6.

#### **6.5 Data Quality Testing**

Data quality testing is the practice of verifying that data held in a platform layer is complete, consistent and correct, and that transformations applied to it have produced the intended result. The principal categories of check applied are:

- Completeness – verifying that all expected data is present and nothing has been lost during processing.
- Consistency – verifying that data is represented uniformly and does not contradict itself.
- Correctness – verifying that values hold the expected content, format and meaning.
- Source-to-target validation – comparing data in a target layer against the corresponding source data.
- Transformation validation – verifying that logic applied between layers produced the intended output.

I applied these checks when performing unit testing of Bronze and Silver layer data during Week 5.

## **6.6 Security and Governance**

Security and governance within a data platform concern the control of access to data and the maintenance of that control over time. Governance establishes the rules under which data may be accessed and by whom; security applies those rules in practice. The principles applied include controlled access so that data is available only to authorised users, access granted according to genuine requirement rather than general availability, recognition that different layers hold data of different sensitivity, protection of raw and intermediate data not curated for general consumption, and maintenance of governance as the platform develops.

I worked on security governance for the Bronze and Silver layers during Week 7. Specific tools, policies, roles and configurations used within the project environment are internal to the organisation and are not described in this report.

## **6.7 Layered Data Architecture**

The layered data architecture is not a single tool but the organising principle of the entire platform. It arranges processing into successive stages – Bronze, Silver, Silver Conformed and Gold – in which each stage has a defined responsibility and produces data of a higher standard than the stage before it. Its principal benefits are:

- Data quality improves progressively rather than in a single unverifiable step.
- Raw data is preserved, allowing processing to be repeated or corrected.
- Issues can be isolated to the layer in which they arise.
- Data can be traced from the final output back to its originating source system.
- New source systems can be introduced by integrating them into the existing structure.

## CHAPTER 7

### PROJECT IMPLEMENTATION / TRAINING DETAILS

#### 7.1 Project Overview

The internship was structured in two phases. The first three weeks consisted of structured technical training intended to establish the foundations required for data engineering work. The remaining five weeks consisted of assignment to the Saluda project and contribution to defined tasks within it.

| Week   | Period               | Activity                                                      |
|--------|----------------------|---------------------------------------------------------------|
| Week 1 | 01 Jul – 07 Jul 2026 | SQL training and introduction to Snowflake                    |
| Week 2 | 08 Jul – 14 Jul 2026 | Snowflake training                                            |
| Week 3 | 15 Jul – 21 Jul 2026 | Databricks training                                           |
| Week 4 | 22 Jul – 28 Jul 2026 | Saluda project assignment and source system analysis          |
| Week 5 | 29 Jul – 04 Aug 2026 | Bronze and Silver layer unit testing (all three sources)      |
| Week 6 | 05 Aug – 11 Aug 2026 | Silver Conformed layer data modelling                         |
| Week 7 | 12 Aug – 18 Aug 2026 | Security governance for the Bronze and Silver layers          |
| Week 8 | 19 Aug – 01 Sep 2026 | Movement of data towards the Silver Conformed and Gold layers |

*Note: The week-ending dates above are indicative and should be adjusted to match the organisation's official internship schedule.*

#### 7.2 Week 1 – SQL and Snowflake

The first week was devoted to training in SQL and to an introduction to Snowflake. The SQL component covered fundamentals and practical querying, together with working with structured data and performing data querying and validation. The emphasis was practical, focusing on using SQL to examine data and confirm whether it meets expectations. This reflects the way SQL is used in data engineering, where in addition to expressing transformation logic it is the primary means by which a data engineer inspects and verifies data. The week also introduced Snowflake and the concepts of cloud data warehousing.

Outcome: I acquired the ability to query and validate structured data using SQL and gained an initial understanding of cloud data warehousing concepts.

#### 7.3 Week 2 – Snowflake

The second week continued with Snowflake training, covering Snowflake concepts, cloud data warehousing, data storage and processing, SQL-based data operations, and how Snowflake can be used within modern data platforms. Building on the SQL foundation of Week 1, this week moved the focus from individual queries to platform thinking: a single query answers a single question,

whereas a data warehouse is an environment in which large volumes of data from across an organisation are held, processed and queried consistently. Understanding this was necessary preparation for working on a platform such as Saluda.

Outcome: I developed an understanding of cloud data warehousing and of how such a platform supports the storage and processing requirements of a modern data platform.

### **7.4 Week 3 – Databricks**

The third week consisted of Databricks training, covering an introduction to Databricks, modern data engineering, data processing, data transformation, and the role of Databricks within data platforms, and completed the training phase of the internship. Where the previous two weeks addressed structured querying and cloud data warehousing, this week addressed the processing and transformation of data as it moves through a pipeline. Data does not move between layers unchanged; at each stage it is processed according to defined logic. The training also placed the technologies in context: a data platform is built not from one tool but from storage, processing and querying capabilities working together.

Outcome: I gained an understanding of Databricks and of the role of data processing and transformation platforms within modern data engineering.

### **7.5 Week 4 – Saluda Project Assignment**

In the fourth week I was assigned to the Saluda project, a data platform building project, marking the transition from training to project work. My activities were understanding the Saluda project and its objectives, understanding the problem being addressed by the platform, analysing the source systems that supply data to it, understanding the source data provided by each system, and understanding how those systems contribute to the overall data platform. The platform draws from QAD, an enterprise resource planning system; SharePoint, a collaboration and document management platform; and Salesforce, a customer relationship management platform.

#### **7.5.1 Importance of Source System Analysis**

Analysing source systems before designing transformations and data models is a necessary first step rather than a preliminary formality. Transformations encode assumptions about the data they receive – which fields are populated, what values they contain, how records are identified. A transformation written before the source data is understood will encode assumptions that may not hold, and when they prove incorrect it must be rewritten along with any downstream work built upon it.

Data models depend equally on knowledge of the sources: a conformed model cannot be designed without knowing which entities exist in which systems. Validation likewise requires a definition of correctness, which the analysis provides. The sources also differ in character – QAD generates structured transactional data automatically, SharePoint holds information that may be maintained manually and vary in completeness, and Salesforce holds customer data likely to overlap with other systems – and recognising this in advance influences how each source is validated and integrated.

Outcome: I developed an understanding of the Saluda project, the problem it addresses, and the three source systems and their contribution to the platform.

## **7.6 Week 5 – Bronze and Silver Layer Unit Testing**

In the fifth week I performed unit testing of Bronze and Silver layer data for all three source systems. Unit testing here means verifying that a specific unit of the pipeline has produced the correct result: rather than testing program code in isolation, it verifies the data the code has produced.

### **7.6.1 Data Validation and Data Quality**

Data validation confirms that data meets defined expectations before it is relied upon. It is required because the successful execution of a process does not guarantee the correctness of its output: a load can complete without error while omitting records, truncating values or applying transformation logic incorrectly. Data quality is the broader property that validation measures – data of good quality is complete, consistent, correct and fit for its purpose – and within a layered platform quality is established progressively, so testing at each layer confirms that this progression has actually occurred.

### **7.6.2 Source-to-Target and Transformation Validation**

Source-to-target validation compares data in a target layer against the corresponding data at its source, confirming that records have not been lost, that values have not been altered unintentionally, and that the target agrees with the source. For Bronze layer testing, close correspondence with the source is expected, and any divergence indicates a problem in ingestion. Transformation validation confirms that logic applied between layers produced the intended result: between Bronze and Silver, data is cleaned and transformed, so the Silver layer is expected to match not the Bronze layer exactly but the result the defined logic should produce when applied to it.

### **7.6.3 Completeness, Consistency and Correctness**

The testing addressed three properties of the data: completeness, that all expected data is present and no records or values were lost; consistency, that data is represented uniformly and does not contradict itself; and correctness, that values hold the expected content, format and meaning. Testing was carried out separately for each source system at both layers, and discrepancies identified were reported for investigation. The specific test cases, volumes validated and results obtained are confidential to the project.

Outcome: I completed unit testing of Bronze and Silver layer data for all three source systems and developed practical understanding of data validation, data quality and source-to-target verification.

## **7.7 Week 6 – Silver Conformed Layer Data Modelling**

In the sixth week I was assigned to work on data modelling for the Silver Conformed layer.

### **7.7.1 Why a Conformed Layer is Required**



Cleaning data does not make data from different systems comparable. After Silver layer processing, data from QAD, SharePoint and Salesforce is structured and reliable, but each set still reflects the design of the system it came from: the same business entity may be identified differently, described using different field names, categorised using different code values and recorded at different levels of detail. If such data were combined directly, records referring to the same entity might not be recognised as related and inconsistent categorisations might be treated as equivalent, so the result would appear integrated without genuinely being so. The conformed layer prevents this by establishing a single agreed representation before integration takes place.

### **7.7.2 Standardisation and Integration**

Standardisation establishes a single way of representing each business entity, so that its representation no longer depends on which system supplied it; without this, a report combining data from three systems would carry three different definitions of the entities it describes. Integration then brings the three sources together within that standardised structure, so the platform holds one integrated view of each entity, while ensuring that relationships between entities remain consistent.

### **7.7.3 Importance of Data Modelling and Position in the Pipeline**

Data modelling is the activity through which the conformed structure is defined, determining which business entities the platform recognises, how each is defined and how they relate. Everything above the conformed layer depends on the model beneath it: a model that reflects business reality accurately makes downstream development straightforward, while one that does not creates persistent difficulty, since corrections at this level require rework across every layer built upon it. The layer's position matters equally – below it the Silver layer holds data that is clean but still source-oriented, above it the Gold layer holds business-oriented data, and without the conformed layer the Gold layer would have to perform standardisation, integration and business preparation at once. The specific entities, structures and relationships defined within the project's model are confidential and are not reproduced here.

Outcome: I contributed to data modelling work for the Silver Conformed layer and developed an understanding of standardisation, integration and the role of data modelling within a layered platform.

## **7.8 Week 7 – Security Governance**

In the seventh week I worked on security governance for the Bronze and Silver layers of the platform.

### **7.8.1 Importance of Security and Access Control**

A data platform concentrates an organisation's information into a single environment, which is what makes it valuable and equally what makes security essential: without deliberate control, access to the platform could amount to access to the organisation's entire data holdings. Access

control determines who may access which data and under what conditions, on the principle that access should be granted according to genuine requirement rather than general availability.

### **7.8.2 Governance and Data Protection**

Governance is the framework within which access decisions are made and maintained. Where access control is the mechanism, governance is the set of rules it enforces and the process by which those rules are established, reviewed and kept current. Governance matters because requirements change over time, so security is a continuing activity rather than a configuration completed once.

### **7.8.3 Security Considerations for Different Data Layers**

Different layers hold data in different states and warrant different treatment. The Bronze layer holds raw ingested data that has not been validated, standardised or curated, so access is generally restricted to those who need to work with raw data directly. The Silver layer holds cleaned and transformed data which, although more usable, remains an intermediate stage rather than a curated business output, and is similarly controlled. Higher layers hold progressively more refined data intended for broader consumption. The general principle is that layers holding raw and intermediate data require the most restrictive controls. Specific security products, policies, roles and configurations used within the project environment are internal to the organisation and are not described in this report.

Outcome: I worked on security governance for the Bronze and Silver layers and developed an understanding of access control, governance and layer-appropriate security within an enterprise data platform.

## **7.9 Week 8 – Silver Conformed and Gold Layers**

In the eighth and final week I worked on moving data from the Silver layer towards the Silver Conformed and Gold layers.

### **7.9.1 Silver to Silver Conformed to Gold**

This progression represents the final stages of the pipeline: data enters in a cleaned but source-oriented state and leaves standardised, integrated and business-oriented. The Silver to Silver Conformed transition applies standardisation and integration, mapping data that still reflects its originating system into the common conformed representation. The Silver Conformed to Gold transition applies business orientation, refining that data according to business requirements to produce datasets intended for reporting and analytical consumption.

### **7.9.2 Data Refinement and Business-Oriented Data**

Data refinement describes the progressive improvement of data as it moves through the platform: Bronze preserves what was received, Silver cleans and structures it, Silver Conformed standardises and integrates it, and Gold organises it for business use. Because Gold layer data has already been validated, standardised and integrated, consumers can use it directly. This ensures consistency: when all consumers draw from the same governed layer, different teams analysing the same subject arrive at the same figures.

### 7.9.3 Scope of My Contribution

A distinction should be drawn between the architecture described above and my personal contribution to it. The account of the Silver to Gold progression and of the role of the Gold layer is a general explanation of the platform's architecture, given so that the context of the work is clear. My own contribution during this week was to work on moving data from the Silver layer towards the Silver Conformed and Gold layers, building on the analysis, validation, modelling and governance work of the preceding weeks.

Outcome: I supported the movement of data towards the Silver Conformed and Gold layers and developed an understanding of how refined, business-oriented data is prepared for downstream consumption.

### 7.10 Overall Project Workflow

The complete workflow, and the position of my contribution within it, is summarised below. My work followed the sequence of the pipeline itself: analysis, then verification, then design, then governance, and finally delivery.

```
Source Systems (QAD, SharePoint, Salesforce)
|--> BRONZE   (ingested)           [unit testing - Wk 5] [governance - Wk 7]
|--> SILVER   (cleaned)            [unit testing - Wk 5] [governance - Wk 7]
|--> SILVER CONFORMED             [data modelling - Wk 6] [movement - Wk 8]
|--> GOLD     (business-ready)     [movement - Wk 8]
|--> Downstream Analytics and Reporting
```

*Figure 2 – Overall Project Workflow and Scope of Internship Contribution*

## **CHAPTER 8**

### **OUTCOMES OF THE PROJECT**

#### **8.1 Key Deliverables**

The following were delivered through my contribution to the Saluda project:

- Completed analysis of the QAD, SharePoint and Salesforce source systems and the data they contribute.
- Unit testing of Bronze layer data for all three source systems.
- Unit testing of Silver layer data for all three source systems.
- Source-to-target and transformation validation covering completeness, consistency and correctness.
- Reporting of discrepancies identified during validation for investigation and resolution.
- Contribution to data modelling work for the Silver Conformed layer.
- Security governance work for the Bronze and Silver layers.
- Support for the movement of data from the Silver layer towards the Silver Conformed and Gold layers.

#### **8.2 Technical Skills Developed**

- SQL – fundamentals and practical querying, applied to structured data and to data validation.
- Snowflake – platform concepts, cloud data warehousing, data storage and processing, and SQL-based data operations.
- Databricks – introduction to the platform and its role in data processing and transformation.
- Data modelling – defining how business entities are represented within a standardised structure.

#### **8.3 Data Engineering Skills**

- Source system analysis – studying source systems before designing transformations and models.
- Layered data architecture – understanding the Bronze, Silver, Silver Conformed and Gold structure.
- Data quality management – applying completeness, consistency and correctness checks.
- Source-to-target and transformation validation across pipeline stages.
- Data standardisation and integration – bringing heterogeneous source data into a common representation.
- Security governance – applying access control principles appropriate to each layer.

#### **8.4 Professional Skills**

- Working within a professional project team on a live client engagement.
- Attention to detail, developed particularly through data validation work.

- Technical communication, through reporting findings and discussing them with the team.
- Handling confidential client data responsibly and observing organisational restrictions.
- Time management across a structured eight-week schedule with defined weekly assignments.

## 8.5 Project Contribution

| Week   | Contribution to the Project                                                                                                                                                                          |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 4 | Analysed the QAD, SharePoint and Salesforce source systems and their contribution to the platform, establishing the understanding required for subsequent work.                                      |
| Week 5 | Performed unit testing of Bronze and Silver layer data for all three source systems, verifying completeness, consistency and correctness and validating source-to-target and transformation results. |
| Week 6 | Contributed to data modelling for the Silver Conformed layer, supporting the standardisation and integration of data from the three source systems.                                                  |
| Week 7 | Worked on security governance for the Bronze and Silver layers, supporting controlled access and data protection at the raw and intermediate stages.                                                 |
| Week 8 | Supported the movement of data from the Silver layer towards the Silver Conformed and Gold layers, contributing to the delivery of business-oriented data.                                           |

## 8.6 Learning Outcomes

1. Understanding of how enterprise data platforms are structured, and why data is processed in successive layers rather than in a single transformation.
2. Understanding of why source systems must be analysed before transformations and data models are designed.
3. Practical understanding that a process completing successfully does not guarantee that its output is correct, and that data must be verified rather than assumed.
4. Understanding of data quality as a measurable property expressed through completeness, consistency and correctness.
5. Understanding of why data from different systems requires standardisation before it can be integrated meaningfully.
6. Understanding of the role of data modelling as the foundation upon which downstream layers depend.
7. Understanding of why security and governance must be designed into a data platform rather than added afterwards, and why raw and intermediate layers require the most restrictive controls.
8. Understanding of how business-oriented data in the Gold layer supports consistent reporting and analytics.
9. Practical experience of applying training in SQL, cloud data warehousing and data processing platforms to a live project.

## **CHAPTER 9**

### **CONCLUSION**

The Saluda project addresses a problem common to organisations that operate multiple independent business systems: data that is valid within each individual system does not, by itself, produce a consistent view of the organisation as a whole. The project resolves this by building a layered data platform that draws data from the QAD, SharePoint and Salesforce source systems and processes it through Bronze, Silver, Silver Conformed and Gold layers, so that data is progressively validated, cleaned, standardised, integrated and prepared for business consumption.

The layered architecture is central to the platform's design. Because each layer has a defined responsibility, quality is established in controlled stages rather than in a single unverifiable step, issues can be isolated to the layer in which they arise, and data can be traced from the final business output back to the source system that supplied it. The Silver Conformed layer occupies a particularly important position, since it is the stage at which data from three different systems is standardised into consistent business entities and made genuinely comparable.

My internship at BI3 Technologies as a Data Engineer Intern, carried out from 01 July 2026 to 01 September 2026, provided practical exposure to this architecture. The first three weeks established the technical foundation through training in SQL, Snowflake and Databricks. The subsequent five weeks applied that foundation to the Saluda project: analysing the three source systems, performing unit testing of Bronze and Silver layer data for each of them, contributing to data modelling for the Silver Conformed layer, working on security governance for the Bronze and Silver layers, and supporting the movement of data towards the Silver Conformed and Gold layers.

A recurring lesson across this work was that reliability in a data platform is achieved deliberately rather than assumed. Data must be validated before it is trusted, since a process may complete without error while producing an incorrect result; data from different systems must be standardised before it is combined; and access must be governed at every layer.

Overall, the internship provided a comprehensive introduction to data engineering as it is practised in industry, demonstrating how modern data platforms are designed, validated, governed and delivered, and how the activities of analysis, testing, modelling and governance combine to produce a platform on which an organisation can rely. The work aligns with SDG 9 - Industry, Innovation and Infrastructure, SDG 8 - Decent Work and Economic Growth, and SDG 4 - Quality Education.

#### **9.1 Future Enhancements**

1. Extension of Testing Coverage – Extend unit testing beyond the Bronze and Silver layers to cover the Silver Conformed and Gold layers, so that validation applies across the full pipeline.
2. Automated Data Quality Checks – Automate validation so that completeness, consistency and correctness checks are executed routinely as part of processing rather than manually.

3. Expansion of the Conformed Data Model – Extend the Silver Conformed model to cover additional business entities as further requirements are identified.
4. Extension of Security Governance – Extend governance controls to the Silver Conformed and Gold layers so that access is governed consistently at every stage.